

Aarya Thakur

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SUMMARY

Final year AI & Data Science student (graduating May 2026) specializing in LLM systems, RAG pipelines, and agentic AI. Proven in high-pressure environments with **2 internships** and **6 hackathon wins**

EDUCATION

Thadomal Shahani Engineering College

Bachelor of Engineering in Artificial Intelligence and Data Science

CGPA 9.43

November 2022 – May 2026

EXPERIENCE

Fullstack Intern

June 2025 – July 2025

Augenblick Consulting

Remote

- Implemented **Deep Zoom Technology**, improving large image rendering speed and enabling seamless web-based navigation of **gigapixel images**.
- Built a **RAG pipeline** ingesting 10,000+ RIS records, converting them into **vector embeddings**, enabling accurate research paper classification, and advanced rule-based filtering.

AI Intern

January 2025 – April 2025

Nimrobo AI

Remote

- Integrated **citation tracking** into AI agent outputs, grounding responses in source documents and enabling users to verify claims directly.
- Developed a **container orchestration system** that automatically spins up isolated AI agent instances with **WebSocket** access, and autonomous lifecycle management.

PROJECTS

MatryoshkaSearch  | *Retrieval Pipeline with Custom Embeddings* | Python, LangGraph, FastAPI, ChromaDB

- Fine-tuned embedding model on ~7,000 financial Q&A pairs using **Matryoshka Representation Learning** achieving **10–13% gains** on NDCG over the base model.
- Built a **LangGraph** retrieval pipeline with **semantic chunking**, query routing and **cross-encoder reranking**, containerized the full stack with Docker.

CollabNext  | *Real-Time Collaborative Editor with AI* | Next.js, TypeScript, Firebase, WebSocket, Yjs

- Built a real-time collaborative content studio with **WebSockets** and **Yjs CRDTs**, ensuring conflict-free synchronization and smooth multi-user editing.
- Developed AI workflows using Gemini that assisted with ghostwriting, story generation, and sentiment analysis.

Metaheuristic Job Scheduler  | *Simulated Annealing & Genetic Algorithms* | Python

- Implemented **Simulated Annealing** to minimize holding costs and delay penalties, and a **Genetic Algorithm** with custom chromosome encoding, crossover, and mutation strategies for superior global exploration.
- Benchmarked **Genetic Algorithm** and **Simulated Annealing** across job sizes up to **5,000 tasks**, GA achieved up to **49% lower scheduling cost**, while SA converged **8.5× faster**.

ACHIEVEMENTS

Hackathons (6 wins): 1st Place: Gajshield KJSSE Hacks 8.0, Hackniche 3.0, Augenblick Hackathon (2025); Runner-up: TSEC Hacks '25, Code Odyssey 3.0, IEEE Techithon (2024–25)

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL

Databases: PostgreSQL, MySQL, Firestore, ChromaDB

AI/ML Tools: HuggingFace, LangChain, Gemini API, LangGraph, PyTorch

Frameworks/Libraries: NumPy, Pandas, Next.js, Django, FastAPI, Node.js, Express.js

Tools/Platforms: Git, GitHub, Docker, Linux, Vercel